

SIMATS ENGINEERING
CO3-ASSESSMENT TOOL 1
Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

Assessment Tool Weightage: 40%

Dr.V.Parthipan

Code of Conduct:

I _M.Greeshma(192524195)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Greeshma

criteria	Understanding of concepts 2.5	Experimental design 3	Use of tools 2	Documentation 1	Total 10
Q1					10
Q2					9.5
Q3					9
Q4					9

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, and ACK). Demonstrate the role of each flag and how they establish a reliable connection.

The image displays a Wireshark packet capture of a TCP connection setup. The packet list shows the following key packets:

No.	Time	Source	Destination	Protocol	Length	Info
2...	2409.2603.10...	2409.2603.10...	2409.2603.10...	TLSv1..	401	Application Data
2...	2409.64:ff9b...	2409.64:ff9b...	2409.64:ff9b...	TCP	86	59948 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1340 WS=256 SACK_PERM=1
2...	2603.2409:40...	2409.64:ff9b...	2409.64:ff9b...	TCP	74	443 → 59947 [ACK] Seq=8560 Ack=527 Win=12583424 Len=0
2...	2603.2409:40...	2409.64:ff9b...	2409.64:ff9b...	TLSv1..	533	Application Data
2...	2603.2409:40...	2409.64:ff9b...	2409.64:ff9b...	TCP	74	443 → 59947 [ACK] Seq=9019 Ack=854 Win=12582912 Len=0
2...	2409.2603:10...	2409.64:ff9b...	2409.64:ff9b...	TCP	74	59947 → 443 [ACK] Seq=854 Ack=9019 Win=65024 Len=0
2...	64:f..2409:40...	2409.64:ff9b...	2409.64:ff9b...	TCP	86	443 → 59948 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1300 WS=256 SACK_PERM=1
2...	2409.64:ff9b...	2409.64:ff9b...	2409.64:ff9b...	TCP	74	59948 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
2...	2409.64:ff9b...	2409.64:ff9b...	2409.64:ff9b...	TLSv1..	303	Client Hello
2...	64:f..2409:40...	2409.64:ff9b...	2409.64:ff9b...	TCP	1354	443 → 59921 [ACK] Seq=1 Ack=7210 Win=16385 Len=1280 [TCP segment of a reassembled PDU]
2...	64:f..2409:40...	2409.64:ff9b...	2409.64:ff9b...	TCP	1354	443 → 59921 [ACK] Seq=1281 Ack=7210 Win=16385 Len=1280 [TCP segment of a reassembled PDU]
2...	2409.64:ff9b...	2409.64:ff9b...	2409.64:ff9b...	TCP	74	59921 → 443 [ACK] Seq=7210 Ack=2561 Win=255 Len=0

Frame 4: 1374 bytes on wire (10992 bits), 1374 bytes captured (10992 bits) on interface \Device\NPF_{243174C0-19DA-447C-86AE...}

Ethernet II, Src: 58:cd:c9:b9:3b:a5 (58:cd:c9:b9:3b:a5), Dst: 86:1f:7d:fa:2a:59 (86:1f:7d:fa:2a:59)

Internet Protocol Version 6, Src: 2409:40f4:411f:538e:a880:a072:817:aab2, Dst: 64:ff9b::14be:918f

Transmission Control Protocol, Src Port: 59921, Dst Port: 443, Seq: 293, Ack: 1, Len: 1300

The Windows command prompt shows the following output:

```
C:\WINDOWS\system32>nslookup example.com
Server: Unknown
Address: 172.16.73.136

Non-authoritative answer:
Name: example.com
Addresses: 2606:4700:9c65:72db:f2fd:d3d:ef6b:ff98
           104.20.23.154
           172.66.147.243

C:\Users\Mettupalle Greeshma>
```

2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability.

The screenshot displays the Wireshark network protocol analyzer interface. The main pane shows a list of captured packets, with the selected packet (No. 26) being a DNS Standard query. The packet details pane shows the structure of the DNS query, including the Ethernet II header, Internet Protocol Version 6 header, and the Domain Name System (query) header. The packet bytes pane shows the raw data of the packet.

Overlaid on the Wireshark window is a Windows command prompt window. The command prompt shows the execution of the `nslookup example.com` command. The output of the command is displayed, showing the server address (172.16.73.136) and the non-authoritative answer for the domain `example.com`, including its IP addresses (104.20.23.154 and 172.66.147.243).

3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark.

Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP.

The image displays a Wireshark packet capture window titled "Wi-Fi". The packet list shows several TCP retransmissions. The packet details pane for the selected packet (No. 66) shows the following information:

- Frame 167: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{243174C0-19DA-447C-86AE-9AAE1A764EA8}, id 0
- Ethernet II, Src: 58:cd:c9:b9:3b:a5 (58:cd:c9:b9:3b:a5), Dst: 86:1f:7d:fa:2a:59 (86:1f:7d:fa:2a:59)
- Internet Protocol Version 4, Src: 172.16.73.135, Dst: 192.178.158.84
- Transmission Control Protocol, Src Port: 56563, Dst Port: 443, Seq: 1301, Ack: 1, Len: 0

The packet bytes pane shows the raw data of the packet:

```
0000 86 1f 7d fa 2a 59 58 cd c9 b9 3b a5 08 00 45 00 ..}.^YX. ..;...E.
0010 00 34 af e6 40 00 00 06 f6 3e ac 10 49 87 c0 b2 .4. @... ->..I...
0020 9e 54 dc f3 01 bb 6f b5 66 51 ce e0 a0 48 00 10 .T...o... fQ...H...
0030 00 ff 21 ef 00 00 01 01 05 0a ce e0 a0 47 ce e0 ..!.....G...
0040 a0 48 -H
```

Overlaid on the bottom right is a terminal window showing a curl command and its output:

```
C:\WINDOWS\system32\cmd.exe
operable program or batch file.

C:\Users\Mettupalle Greeshma>curl http://example.com
<!doctype html><html lang="en"><head><title>Example Domain</title><link rel="icon" href="data:;"/><meta name="viewport" content="width=device-width, initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</style></head><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without needing permission. Avoid use in operations.</p><p><a href="https://iana.org/domains/example">Learn more</a></p></div></body></html>

C:\Users\Mettupalle Greeshma>
```

4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP.

The screenshot displays the Wireshark network protocol analyzer interface. The packet list pane on the left shows a series of DNS queries and responses. The packet details pane on the right shows the structure of a DNS query, including the Ethernet II header, Internet Protocol Version 4 header, and User Datagram Protocol header. The packet bytes pane at the bottom shows the raw data of the captured packet.

Wireshark Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
2...	2409.2409.40...	172.16.73.136	172.16.73.136	DNS	359	Standard query response 0x93c9 AAAA www.bing.com CNAME www-wm.bing.com.trafficmanager.net CNAME www.bing.com.edgekey.net CNAME e86303.dscx.akamaiedge.net AAAA 2600:14...
2...	2409.2409.40...	172.16.73.136	172.16.73.136	DNS	339	Standard query response 0x93c9 AAAA www.bing.com CNAME www-wm.bing.com.trafficmanager.net CNAME www.bing.com.edgekey.net CNAME e86303.dscx.akamaiedge.net AAAA 2600:14...
2...	2409.2409.40...	172.16.73.136	172.16.73.136	DNS	363	Standard query response 0x89ab A www.bing.com CNAME www-wm.bing.com.trafficmanager.net CNAME www.bing.com.edgekey.net CNAME e86303.dscx.akamaiedge.net A 23.44.10.66 A...
2...	172.16.73.136	172.16.73.136	172.16.73.136	DNS	343	Standard query response 0x89ab A www.bing.com CNAME www-wm.bing.com.trafficmanager.net CNAME www.bing.com.edgekey.net CNAME e86303.dscx.akamaiedge.net A 23.44.10.75 A...
3...	172.16.73.136	172.16.73.136	172.16.73.136	DNS	113	Standard query 0x746e A 51b826ae1d11304f650e5bfff5846d36.azr.footprintdns.com
3...	172.16.73.136	172.16.73.136	172.16.73.136	DNS	113	Standard query 0x2838 AAAA 51b826ae1d11304f650e5bfff5846d36.azr.footprintdns.com
3...	2409.2409.40...	172.16.73.136	172.16.73.136	DNS	133	Standard query 0x2838 AAAA 51b826ae1d11304f650e5bfff5846d36.azr.footprintdns.com
3...	2409.2409.40...	172.16.73.136	172.16.73.136	DNS	133	Standard query 0x746e A 51b826ae1d11304f650e5bfff5846d36.azr.footprintdns.com
4...	172.16.73.136	172.16.73.136	172.16.73.136	DNS	113	Standard query 0x2838 AAAA 51b826ae1d11304f650e5bfff5846d36.azr.footprintdns.com
4...	172.16.73.136	172.16.73.136	172.16.73.136	DNS	113	Standard query 0x746e A 51b826ae1d11304f650e5bfff5846d36.azr.footprintdns.com
5...	172.16.73.136	172.16.73.136	172.16.73.136	DNS	116	Standard query 0x24ea HTTPS 201295d1a9eae6d3ed24f1bd4fe9c5d2.canary.fp.wan.azure.com
5...	172.16.73.136	172.16.73.136	172.16.73.136	DNS	116	Standard query 0x45c6 AAAA 201295d1a9eae6d3ed24f1bd4fe9c5d2.canary.fp.wan.azure.com

Wireshark Packet Details:

- Frame 154: 116 bytes on wire (928 bits), 116 bytes captured (928 bits) on interface \Device\NPF_{243174C0-19DA-447C-86AE-9A6E1A764EA8}, id 0
- Ethernet II, Src: 58:cd:c9:b9:3b:a5 (58:cd:c9:b9:3b:a5), Dst: 86:1f:7d:fa:2a:59 (86:1f:7d:fa:2a:59)
- Internet Protocol Version 4, Src: 172.16.73.135, Dst: 172.16.73.136
- User Datagram Protocol, Src Port: 57865, Dst Port: 53
- Domain Name System (query)

Windows Command Prompt:

```
C:\Users\Mettupalle Greeshma>nslookup google.com
Server:      Unknown
Address:     172.16.73.136

Non-authoritative answer:
Name:       google.com
Addresses:  2404:6800:4002:829::200e
            192.178.158.102
            192.178.158.139
            192.178.158.100
            192.178.158.101
            192.178.158.113
            192.178.158.138
```